

因果推断: 作业合集

Peng Ding, A First Course in Causal Inference

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整理说明: 本合集根据附件中的 Problem Set 1-6 题单、仓库中已有的 homework4.tex-homework6.tex、Peng Ding 教材源文件与本地中文解答整理。教材题按 “Ding 章. 题” 标号; 课堂附加题按 “PS 编号” 标号。若数据文件不在本地, 则给出可复现的分析代码框架, 不伪造数值结论。

第 1 次作业

题 1 — Ding 2.1: perfect doctor 例子

来源: 教材 Chapter 2, Problem 2.1; 附件 HW1 的 Exercise 1。

设

$$Y(0) \sim N(0, 1), \quad \tau = -0.5 + Y(0), \quad Y(1) = Y(0) + \tau.$$

医生按 $Z = \mathbb{1}(\tau \geq 0)$ 分配治疗, 观测结果为 $Y = ZY(1) + (1 - Z)Y(0)$ 。求

$$E(Y | Z = 1) - E(Y | Z = 0).$$

Answer

令 $X = Y(0)$ 。因为 $\tau = -0.5 + X$, 所以 $Z = 1$ 当且仅当 $X \geq 0.5$ 。又 $Y(1) = 2X - 0.5$, 因此

$$E(Y | Z = 1) - E(Y | Z = 0) = E(2X - 0.5 | X \geq 0.5) - E(X | X < 0.5).$$

标准正态的截断均值为

$$E(X | X \geq a) = \frac{\phi(a)}{1 - \Phi(a)}, \quad E(X | X < a) = -\frac{\phi(a)}{\Phi(a)}.$$

取 $a = 0.5$ 得

$$2 \frac{\phi(0.5)}{1 - \Phi(0.5)} - 0.5 + \frac{\phi(0.5)}{\Phi(0.5)} \approx 2.292.$$

这与真实平均因果效应 $E(\tau) = -0.5$ 方向相反。原因不是代数错误, 而是分配机制直接依赖个体治疗效应; 处理组本来就是潜在结果和治疗效应较高的人。

题 2 — Ding 3.4: 二元结果 CRE 下的 Fisher 精确检验

来源: 教材 Chapter 3, Problem 3.4; 附件 HW1 的 Exercise 3。

在完全随机实验中, 二元结果的 2×2 表为

	$Y = 1$	$Y = 0$	total
$Z = 1$	n_{11}	n_{10}	n_1
$Z = 0$	n_{01}	n_{00}	n_0
total	m_1	m_0	n

在 Fisher sharp null H_{0F} 下, 说明任意统计量 $T(n_{11}, n_{10}, n_{01}, n_{00})$ 都是 n_{11} 与固定常数的函数, 并给出 n_{11} 的精确分布。

Answer

在 H_{0F} 下, 每个单位在 treatment 和 control 下的潜在结果相同。因此所有观测到的 Y 值在重随机化中保持不变, 固定的量包括

$$n, \quad n_1, \quad n_0, \quad m_1 = n_{11} + n_{01}, \quad m_0 = n_{10} + n_{00}.$$

一旦 n_{11} 给定, 其余三个格子都由边际和确定:

$$n_{10} = n_1 - n_{11}, \quad n_{01} = m_1 - n_{11}, \quad n_{00} = n_0 - m_1 + n_{11}.$$

所以任何 $T(n_{11}, n_{10}, n_{01}, n_{00})$ 都可写成 n_{11} 与固定边际和的函数。

在完全随机化下, n_1 个处理名额是从 n 个单位中无放回抽取的, 而其中有 m_1 个单位的固定结果为 1。所以

$$n_{11} \sim \text{Hypergeometric}(N = n, K = m_1, r = n_1),$$

即

$$\Pr(n_{11} = k) = \frac{\binom{m_1}{k} \binom{n-m_1}{n_1-k}}{\binom{n}{n_1}},$$

其中 k 必须使四个格子均非负。这正是 Fisher exact test 与 CRE 下 FRT 的等价来源。

题 3 — Ding 3.5: lady tasting tea

来源: 教材 Chapter 3, Problem 3.5; 附件 HW1 的 Exercise 5。

Lady tasting tea 实验中共有 8 杯茶, 真实有 4 杯 milk first。女士也选择 4 杯为 milk first。令 X 为正确分类的 milk-first 杯数。计算 $\Pr(X = k)$, $k = 0, 1, 2, 3, 4$ 。

Answer

在不能分辨的 sharp null 下, 女士选择的 4 杯等概率地来自全部 $\binom{8}{4} = 70$ 种选择。若 $X = k$, 则她从真实 milk-first 的 4 杯中选中 k 杯, 并从真实 tea-first 的 4 杯中误选 $4 - k$ 杯。因此

$$\Pr(X = k) = \frac{\binom{4}{k} \binom{4}{4-k}}{\binom{8}{4}} = \frac{\binom{4}{k}^2}{70}, \quad k = 0, 1, 2, 3, 4.$$

数值为

$$\Pr(X = 0, 1, 2, 3, 4) = \frac{1, 16, 36, 16, 1}{70}.$$

真实实验 $X = 4$ 的单侧极端概率为 $1/70 \approx 0.0143$, 这就是 Fisher 用来说明随机化检验力量的经典例子。

题 4 — Ding 3.6: 协变量调整的 FRT

来源: 教材 Chapter 3, Problem 3.6; 附件 HW1 的 Exercise 7。

解释两种协变量调整 FRT 为什么是 finite-sample exact: 先把 Y 对 X 回归取残差作为 pseudo outcome, 再定义统计量; 或直接把 Y 对 Z, X 回归中 Z 的系数作为统计量。

Answer

FRT 的 exactness 只要求两件事: sharp null 下所有潜在结果可由观测结果恢复; 统计量在每个随机化分配向量下都是预先定义好的函数。协变量 X 在随机化中固定, 且在 H_{0F} 下 $Y_i(1) = Y_i(0) = Y_i^{\text{obs}}$, 所以每个候选分配 $\mathbf{z}$ 下的 imputed outcome 都相同。

Pseudo-outcome 策略中, 先用固定的 Y^{obs} 和固定的 X 拟合模型, 残差也是固定向量; 随后把处理标签重新随机化并计算残差上的统计量。Model-output 策略中, 对每个候选 $\mathbf{z}$ 拟合同一个形式的回归 $Y^{\text{obs}} \sim \mathbf{z} + X$, 取 $\mathbf{z}$ 的系数。这两个统计量都只是随机化分配 $\mathbf{z}$ 与固定数据 (Y^{obs}, X) 的函数。因此其随机化分布正是实验设计诱导的精确分布, p 值 finite-sample exact。

题 5 — PS1-2: LaLonde 观测数据的变量选择与 Simpson 悖论

来源: Problem Set 1, Problem 2; 数据文件名为 `nsw_cps.dta`。

原题:

枚举 8 个协变量的 $2^8 = 256$ 个子集, 对 `re78` 回归 `treat` 与协变量, 统计 treatment 系数正负、显著性, 并讨论与 Simpson 悖论的关系。具体要求是:

1. 对每个协变量子集拟合一个线性回归模型, 以 `re78` 为结果, 以 `treat` 和该子集为解释变量。
2. 汇总 treatment 系数的符号、显著性以及随协变量集合变化的模式。
3. 结合观测研究中的混杂调整, 讨论这些变化与 Yule-Simpson 悖论的关系。

Answer

本地仓库没有找到附件题单中指定的 `nsw_cps.dta`, 所以这里不给出伪造数值。可复现分析应按下面流程运行:

```
library(haven)
dat <- read_dta("nsw_cps.dta")

outcome <- "re78"
treat <- "treat"
covars <- c("age", "educ", "black", "hispan",
            "married", "nodegree", "re74", "re75")

subsets <- unlist(lapply(0:length(covars), function(k) {
  combn(covars, k, simplify = FALSE)
}), recursive = FALSE)

fit_one <- function(xs) {
  rhs <- c(treat, xs)
  form <- as.formula(paste(outcome, "~", paste(rhs, collapse = " + ")))
  tab <- summary(lm(form, data = dat))$coef
  c(estimate = tab[treat, "Estimate"],
    se = tab[treat, "Std. Error"],
    p = tab[treat, "Pr(>|t|)"],
    k = length(xs))
}

res <- as.data.frame(do.call(rbind, lapply(subsets, fit_one)))
with(res, table(sign = estimate > 0, significant = p < 0.05))
```

```
summary(res$estimate)
plot(res$k, res$estimate); abline(h = 0, lty = 2)
```

解释时应强调：这些回归不是 256 个不同的实验，而是同一份观测数据在不同调整集合下的关联估计。若 treatment 系数随协变量选择发生符号变化，说明 marginal association 与 conditional association 可能方向不同；这正是 Yule-Simpson 悖论在回归语境中的影子。因果上，协变量集合必须由识别假设和设计理由决定，而不能事后凭显著性筛选。

第 2 次作业

题 6 — Ding 4.1: Lemma 4.1 的方差恒等式

来源: 教材 Chapter 4, Problem 4.1; 附件 HW2。

原题:

Prove Lemma 4.1, namely the finite-population covariance identity 证明有限总体恒等式

$$2S(1, 0) = S^2(1) + S^2(0) - S^2(\tau).$$

Answer

令 $\tau_i = Y_i(1) - Y_i(0)$, $\tau = \bar{Y}(1) - \bar{Y}(0)$ 。则

$$\tau_i - \tau = \{Y_i(1) - \bar{Y}(1)\} - \{Y_i(0) - \bar{Y}(0)\}.$$

平方并对 i 求和除以 $n - 1$:

$$S^2(\tau) = S^2(1) + S^2(0) - 2S(1, 0).$$

移项即得结论。这个恒等式说明, 不可观测的潜在结果协方差与个体效应异质性是同一枚硬币的两面。

题 7 — Ding 4.2–4.3: Neyman 方差与 OLS 实现

来源: 教材 Chapter 4, Problems 4.2–4.3; 附件 HW2。

原题:

1. Ding 4.2. Under a CRE, calculate

$$\text{var}\{\hat{Y}(1)\}, \quad \text{var}\{\hat{Y}(0)\}, \quad \text{cov}\{\hat{Y}(1), \hat{Y}(0)\},$$

and use these formulas to calculate $\text{var}(\hat{\tau})$.

2. Ding 4.3. Prove the OLS identities for Neymanian inference in the no-covariate regression of Y on an intercept and Z . Moreover, prove that the HC2 variant of the EHW robust variance estimator recovers the Neyman conservative variance estimator $\hat{V}$ exactly.

Answer

CRE 下, 处理组均值是从有限总体 $\{Y_i(1)\}$ 中抽取 n_1 个的简单随机样本均值, 对照组类似。因此

$$\text{var}\{\hat{Y}(1)\} = \frac{n_0}{nn_1} S^2(1), \quad \text{var}\{\hat{Y}(0)\} = \frac{n_1}{nn_0} S^2(0),$$

且互补抽样给出

$$\text{cov}\{\hat{Y}(1), \hat{Y}(0)\} = -\frac{1}{n}S(1, 0).$$

于是

$$\text{var}(\hat{\tau}) = \frac{n_0}{nn_1}S^2(1) + \frac{n_1}{nn_0}S^2(0) + \frac{2}{n}S(1, 0).$$

代入上一题的恒等式, 可化为

$$\text{var}(\hat{\tau}) = \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n}.$$

对回归 $Y_i = \alpha + \beta Z_i + \varepsilon_i$, 带截距的 OLS 正规方程要求两组内残差和为零, 所以 $\hat{\alpha} = \hat{Y}(0)$, $\hat{\alpha} + \hat{\beta} = \hat{Y}(1)$, 从而

$$\hat{\beta} = \hat{Y}(1) - \hat{Y}(0) = \hat{\tau}.$$

系数相同不代表常规同方差标准误正确; 与 Neyman 保守方差一致的是合适的异方差稳健标准误 (如二组设计中的 HC2 形式)。

题 8 — Ding 4.6: 向量版本的 Neyman 定理

来源: 教材 Chapter 4, Problem 4.6; 附件 HW2。

原题:

The classic result of Neyman (1923) is about a scalar outcome. Extend the potential outcomes to vectors $\mathbf{V}_i(1), \mathbf{V}_i(0) \in \mathbb{R}^K$ and define

$$\tau_{\mathbf{V}} = n^{-1} \sum_i \{\mathbf{V}_i(1) - \mathbf{V}_i(0)\}.$$

The Neyman-type estimator is

$$\hat{\tau}_{\mathbf{V}} = \bar{\mathbf{V}}_1 - \bar{\mathbf{V}}_0 = n_1^{-1} \sum_i Z_i \mathbf{V}_i - n_0^{-1} \sum_i (1 - Z_i) \mathbf{V}_i.$$

Under a CRE, show that $\hat{\tau}_{\mathbf{V}}$ is unbiased for $\tau_{\mathbf{V}}$, find its covariance matrix, and give a possibly conservative covariance estimator.

Answer

设 $Y_i(z) \in \mathbb{R}^K$, $\tau_i = Y_i(1) - Y_i(0)$, 定义有限总体协方差矩阵

$$S_z = (n-1)^{-1} \sum_i \{Y_i(z) - \bar{Y}(z)\} \{Y_i(z) - \bar{Y}(z)\}^{\top},$$

$$S_{\tau} = (n-1)^{-1} \sum_i (\tau_i - \tau)(\tau_i - \tau)^{\top}.$$

逐坐标应用标量 Neyman 定理, 并把协方差项排列成矩阵, 可得

$$\text{Cov}(\hat{\tau}) = \frac{S_1}{n_1} + \frac{S_0}{n_0} - \frac{S_{\tau}}{n}.$$

样本模拟量

$$\hat{V} = \frac{s_1}{n_1} + \frac{s_0}{n_0}$$

在 Loewner order 意义下保守, 因为 S_τ/n 半正定且不可观测。

题 9 — Ding 5.4 与 5.9: SRE 相关推断

来源: 教材 Chapter 5, Problems 5.4, 5.9; 附件 HW2。

原题:

1. Ding 5.4. Compare the CRE and SRE by proving the displayed identity relating the CRE estimator, the SRE estimator, and the imbalance of stratum composition in the textbook.
2. Ding 5.9. Re-analyze the LaLonde data used in the Neyman variance example. Conduct both Fisherian and Neymanian inferences. The original experiment is a CRE; then pretend it is an SRE stratified by race (black, Hispanic, or other), by marital status, and by high-school diploma indicator. Compare these analyses with the CRE analysis.

Answer

若第 k 层大小为 n_k , 权重 $\pi_k = n_k/n$, 层内差均值为

$$\hat{\tau}_k = \bar{Y}_{k1}^{\text{obs}} - \bar{Y}_{k0}^{\text{obs}},$$

则 SRE 的自然估计量是

$$\hat{\tau}_{\text{sre}} = \sum_k \pi_k \hat{\tau}_k.$$

层内按 CRE 计算 Neyman 保守方差, 再加权求和:

$$\hat{V}_{\text{sre}} = \sum_k \pi_k^2 \left\{ \frac{s_k^2(1)}{n_{k1}} + \frac{s_k^2(0)}{n_{k0}} \right\}.$$

LaLonde 再分析题的重点不是套公式, 而是比较不同分层变量下 Fisherian 与 Neymanian 推断的稳定性。本地没有找到题单要求的原始数据文件, 因此应报告可复现框架、分层变量、统计量、随机化机制与结论敏感性, 而不虚构 p 值。

题 10 — Ding 6.6–6.10: 回归调整与协变量调整

来源: 教材 Chapter 6, Problems 6.6–6.10; 6.6 与 6.10 为选做。

原题:

1. Ding 6.6 (选做). Show that the variance of the linearly adjusted estimator decomposes as

$$\text{var}\{\hat{\tau}(\beta_1, \beta_0)\} = \text{var}\{\hat{\tau}(\tilde{\beta}_1, \tilde{\beta}_0)\} + \text{var}\{\hat{\tau}(\beta_1, \beta_0) - \hat{\tau}(\tilde{\beta}_1, \tilde{\beta}_0)\},$$

where $\tilde{\beta}_1, \tilde{\beta}_0$ are the finite-population OLS projection coefficients of $Y_i(1)$ and $Y_i(0)$ on $(1, X_i)$.

2. Ding 6.7. Prove Proposition 6.1 on Lin's estimator for covariate adjustment.
3. Ding 6.8. Prove the textbook formulas for the predictive estimator and the projective estimator.

4. Ding 6.10 (选做) . Show the ANCOVA imbalance representation

$$\hat{\tau}_F = \hat{\tau} - \hat{\gamma}_F^\top \hat{\tau}_X,$$

where $\hat{\gamma}_F$ is the coefficient of X_i in the OLS fit of Y_i on $(1, Z_i, X_i)$.

Answer

CRE 中使用协变量调整的关键原则是：点估计可以用回归实现，但解释仍应回到随机化。Lin 型完全交互回归

$$Y_i \sim 1 + Z_i + X_i + Z_i(X_i - \bar{X})$$

给出渐近上不劣于未调整差均值的估计量；不含交互的 ANCOVA 也常能改善效率，但在存在处理效应异质性时，完全交互形式更稳健。

Predictive estimator 先为每个单位预测两个潜在结果再平均；projective estimator 则把潜在结果线性投影到协变量上以解释效率提升。两者的共同直觉是：协变量只用于解释 outcome variation，不改变随机化保证的无偏性。若分析时忽略交互或使用非线性模型，目标量可能不再是简单 ACE，因此需要额外说明模型估计量对应的 estimand。

题 11 — PS2-2: 把 SRE 当作 CRE 分析

来源：Problem Set 2, Problem 2。

原题：

若真实设计是 SRE，但分析时忽略分层，写出估计量、与分层估计量比较，并讨论效率。

1. Write down the difference-in-means estimator that ignores the stratification.
2. Compare this estimator with the stratified estimator. Are they equal in general? Under what condition would they coincide?
3. Derive or state the variance of both estimators under the SRE design. Which estimator is more efficient?

Answer

忽略分层的差均值为

$$\hat{\tau}_{\text{cre}} = \bar{Y}_{Z=1}^{\text{obs}} - \bar{Y}_{Z=0}^{\text{obs}}.$$

分层估计量为

$$\hat{\tau}_{\text{sre}} = \sum_k \frac{n_k}{n} \left(\bar{Y}_{k,Z=1}^{\text{obs}} - \bar{Y}_{k,Z=0}^{\text{obs}} \right).$$

二者一般不相等，因为未分层差均值隐含的层权重由实际处理组/对照组的层构成决定；只有当每层处理比例 n_{k1}/n_k 都等于总体处理比例 n_1/n ，或层均值结构恰好使加权差异抵消时，二者才一致。

在 SRE 设计下，分层估计量利用了设计信息，方差通常更小或至少不差。忽略分层不会自动产生偏差，但会丢掉设计带来的精度；标准误若按 CRE 写，还可能与真实随机化机制不匹配。

题 12 — PS2-3: 随机实验中用 GLM 估计 ACE

来源: Problem Set 2, Problem 3。

原题:

比较线性回归与 logistic 回归随着协变量逐步加入时对 treatment effect 的估计。题目给出一个 $n = 10000$ 的随机实验模拟: 8 个独立标准正态协变量 $X_i = (X_{i1}, \dots, X_{i8})$; 二元潜在结果按 logistic 模型生成; $Z_i \sim \text{Bernoulli}(0.5)$; 观测结果为 $Y_i = Z_i Y_i(1) + (1 - Z_i) Y_i(0)$ 。分析时不能使用潜在结果。

1. 用线性回归依次拟合 Y_i 对 Z_i 、 $Z_i + X_{i1}$ 、 $\dots$ 、 $Z_i + X_{i1} + \dots + X_{i8}$ 的 9 个模型, 报告 Z 的系数并作图。
2. 用 logistic 回归做同样分析; 也可尝试 probit、tobit、complementary log-log 等链接函数。
3. 讨论分析发现。

Answer

随机实验中, 差均值与线性回归 treatment 系数直接估计 ACE。即使 Y 是二元变量, 线性回归的 treatment 系数仍有清楚的风险差解释。Logistic 回归估计的是条件 log odds ratio; 由于 odds ratio 不可折叠, 加入与 treatment 独立的预测性协变量也可能改变 treatment 系数。因此 logistic 系数随协变量数增加而变化, 不必解释为 confounding 被逐步消除。

可复现分析框架如下:

```
fits_lm <- lapply(0:8, function(k) {
  xs <- paste0("X", seq_len(k))
  rhs <- paste(c("Z", xs), collapse = " + ")
  coef(lm(as.formula(paste("Y ~", rhs)), data = dat))["Z"]
})

fits_glm <- lapply(0:8, function(k) {
  xs <- paste0("X", seq_len(k))
  rhs <- paste(c("Z", xs), collapse = " + ")
  coef(glm(as.formula(paste("Y ~", rhs)),
    data = dat, family = binomial()))["Z"]
})
```

报告时应把线性模型的风险差估计和 logistic 模型的 log odds ratio 分开, 不要把二者画在同一量纲上直接比较。

题 13 — PS2-4 (选做): collapsibility

来源: Problem Set 2, Problem 4。

原题:

解释线性回归风险差与 logistic odds ratio 的 collapsibility 差异。

1. 在线性条件均值模型 $E(Y_i | Z_i, X_i) = \beta_0 + \beta_Z Z_i + \beta_X^\top X_i$ 下, 证明 $\beta_Z = E(Y_i | Z_i = 1, X_i) - E(Y_i | Z_i = 0, X_i)$ 。
2. 在 logistic 模型 $\text{logit Pr}(Y_i = 1 | Z_i, X_i) = \beta_0 + \beta_Z Z_i + \beta_X^\top X_i$ 下, 证明 β_Z 是条件 odds ratio 的对数。

3. 若 $RD(X_{i1}, X_{i2}) = c$ 为常数, 能否推出 $RD(X_{i1})$ 也是常数? 若 Z_i 与 X_i 独立呢?
4. 若 $OR(X_{i1}, X_{i2}) = c$ 为常数, 能否推出 $OR(X_{i1})$ 也是常数? 解释原因。
5. 若 $OR(X_{i1}, X_{i2}) = c$, 且 X_{i1}, X_{i2} 相互独立并且也与 Z_i 独立, $OR(X_{i1})$ 与 c 有什么关系?

Answer

若

$$E(Y | Z, X) = \beta_0 + \beta_Z Z + \beta_X^\top X,$$

则

$$E(Y | Z = 1, X) - E(Y | Z = 0, X) = \beta_Z.$$

风险差在适当独立性条件下具有 collapsibility: 若更细分的风险差 $RD(X_1, X_2) = c$ 为常数, 且 $Z \perp\!\!\!\perp X_2 | X_1$ 或足够强的独立条件成立, 则较粗分层的 $RD(X_1)$ 仍为 c 。

若

$$\text{logit Pr}(Y = 1 | Z, X) = \beta_0 + \beta_Z Z + \beta_X^\top X,$$

则

$$\beta_Z = \log \frac{\Pr(Y = 1 | Z = 1, X) \Pr(Y = 0 | Z = 0, X)}{\Pr(Y = 0 | Z = 1, X) \Pr(Y = 1 | Z = 0, X)}.$$

Odds ratio 一般不可折叠: 即使 $OR(X_1, X_2) = c$ 且 Z 与协变量独立, 边际或较粗分层 $OR(X_1)$ 也通常不等于 c 。这是 logistic treatment 系数在 randomized experiments 中随协变量加入而改变的重要原因, 不应自动解读为偏差减少。

第 3 次作业

题 14 — Ding 10.2–10.3: 观测研究中的识别与 logistic outcome imputation

来源: 教材 Chapter 10, Problems 10.2–10.3; 附件 HW3。

原题:

1. Ding 10.2. Under ignorability and overlap, show that several causal quantities beyond the ACE are nonparametrically identifiable, including the marginal distributions or survival functions of $Y(1)$ and $Y(0)$ and the quantile causal effect

$$\text{QCE}_q = \text{quantile}_q\{Y(1)\} - \text{quantile}_q\{Y(0)\}.$$

Explain why the quantile of the individual causal effect,

$$\tau_q = \text{quantile}_q\{Y(1) - Y(0)\},$$

is not identified by the two marginal potential-outcome distributions alone.

2. Ding 10.3. For a binary outcome, suppose the fully interacted logistic model is

$$E(Y | Z, X) = \Pr(Y = 1 | Z, X) = \frac{\exp(\beta_0 + \beta_z Z + \beta_x^\top X + \beta_{xz}^\top X Z)}{1 + \exp(\beta_0 + \beta_z Z + \beta_x^\top X + \beta_{xz}^\top X Z)}.$$

What is the corresponding outcome-regression estimator for the average causal effect?

Answer

若

$$Z \perp\!\!\!\perp \{Y(1), Y(0)\} | X, \quad 0 < e(X) < 1,$$

则对任何可积函数 g ,

$$E\{g(Y(1), X)\} = E\{E(g(Y) | Z = 1, X)\},$$

$$E\{g(Y(0), X)\} = E\{E(g(Y) | Z = 0, X)\}.$$

因此分布函数、分位数、均值和分层均值都可由条件 outcome distribution 识别。

若 Y 为二元, 并对 $z = 0, 1$ 分别拟合完全交互 logistic model

$$\text{logit } \Pr(Y = 1 | Z = z, X) = \alpha_z + \gamma_z^\top X,$$

则估计

$$\hat{\mu}_z(X_i) = \text{expit}(\hat{\alpha}_z + \hat{\gamma}_z^\top X_i), \quad \hat{\tau}_{oi} = n^{-1} \sum_i \{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\}.$$

完全交互的含义是 treatment 与协变量的关系不被强制相同; 它把两个 treatment arms 的 outcome models 分开估计。

题 15 — Ding 11.1, 11.5, 11.6: propensity score

来源: 教材 Chapter 11, Problems 11.1, 11.5, 11.6; 附件 HW3。

原题:

1. Ding 11.1. Prove another version of the propensity-score dimension-reduction theorem:

$$Z \perp\!\!\!\perp \{Y(1), Y(0), X\} \mid e\{X, Y(1), Y(0)\}.$$

This result does not require strong ignorability and treats $e\{X, Y(1), Y(0)\}$ as a principal unobserved covariate.

2. Ding 11.5. Let $b(X)$ be a balancing score, meaning $Z \perp\!\!\!\perp X \mid b(X)$. Prove the theorem that $b(X)$ is a balancing score if and only if it is finer than the propensity score, i.e. $e(X) = f\{b(X)\}$ for some function f .
3. Ding 11.6. In an observational study with $X = (X_1, X_2)$ and binary subgroup indicator X_1 , the target is

$$\tau(x_1) = E\{Y(1) - Y(0) \mid X_1 = x_1\}, \quad x_1 = 0, 1.$$

Show the IPW identification formula

$$\tau(x_1) = E \left\{ \frac{\mathbb{1}(X_1 = x_1)ZY}{e(X)} - \frac{\mathbb{1}(X_1 = x_1)(1 - Z)Y}{1 - e(X)} \right\} / \Pr(X_1 = x_1),$$

and give the corresponding HT and Hajek estimators.

Answer

若 $e(X) = \Pr(Z = 1 \mid X)$, 则

$$Z \perp\!\!\!\perp X \mid e(X).$$

证明只需注意, 对任意 x 满足 $e(x) = e$,

$$\begin{aligned} \Pr(X = x \mid Z = 1, e(X) = e) &= \frac{\Pr(Z = 1 \mid X = x) \Pr(X = x \mid e(X) = e)}{\Pr(Z = 1 \mid e(X) = e)} \\ &= \Pr(X = x \mid e(X) = e). \end{aligned}$$

若 ignorability $Z \perp\!\!\!\perp \{Y(1), Y(0)\} \mid X$ 成立, 则由于 $e(X)$ 是 balancing score, 在适当正则条件下也可得到

$$Z \perp\!\!\!\perp \{Y(1), Y(0)\} \mid e(X).$$

Chapter 11.5 的要点是: balancing score $b(X)$ 若满足 $Z \perp\!\!\!\perp X \mid b(X)$, 则 propensity score 是最粗的 balancing score, 且任何 balancing score 都蕴含 $Z \perp\!\!\!\perp \{Y(1), Y(0)\} \mid b(X)$ 。Subgroup effects 则提醒我们: 估计 $\tau(X)$ 或 $E\{\tau(X) \mid G(X) = g\}$ 时, overlap 必须在相应子群内检查, 而不是只看总体。

题 16 — Ding 12.2: doubly robust estimator 的替代表达

来源: 教材 Chapter 12, Problem 12.2; 附件 HW3。

原题:

Motivated by the textbook explanation of IPW efficiency, consider the alternative population form

$$\tilde{\mu}_1^{\text{dr2}} = \frac{E[Z\{Y - \mu_1(X, \beta_1)\}/e(X, \alpha)]}{E\{Z/e(X, \alpha)\}} + E\{\mu_1(X, \beta_1)\}.$$

Show that $\tilde{\mu}_1^{\text{dr2}} = \mu_1$ if either $e(X, \alpha) = e(X)$ or $\mu_1(X, \beta_1) = \mu_1(X)$. Give the analogous formula for μ_0 , and give the sample analog of the doubly robust estimator for τ .

Answer

记 $\mu_z(X) = E(Y | Z = z, X)$, $e(X) = \Pr(Z = 1 | X)$ 。ACE 的 DR 估计目标为

$$\tau = E\{Y(1) - Y(0)\}.$$

其总体形式为

$$\begin{aligned}\psi_1 &= \frac{ZY}{e(X)} - \left\{ \frac{Z}{e(X)} - 1 \right\} \mu_1(X), \\ \psi_0 &= \frac{(1-Z)Y}{1-e(X)} - \left\{ \frac{1-Z}{1-e(X)} - 1 \right\} \mu_0(X), \\ \tau &= E(\psi_1 - \psi_0).\end{aligned}$$

若 propensity score 正确, 则对 X 条件化后, 残差项加权期望为零, 所以 outcome model 即使错也不影响一致性。若 outcome model 正确, 则 $E(Y - \mu_Z(X) | Z, X) = 0$, propensity model 即使错也不影响一致性。这就是 doubly robust 的含义: 两个 nuisance components 至少一个正确即可。

题 17 — PS3-2: ATT 的识别、估计与 DR 形式

来源: Problem Set 3, Problem 2。

原题:

在 $Y(0) \perp\!\!\!\perp Z | X$ 下识别 $\text{att} = E\{Y(1) - Y(0) | Z = 1\}$, 并给出 outcome regression、IPW 与 DR 估计。

1. Show that $\text{att} = E(Y | Z = 1) - E\{E(Y | Z = 0, X) | Z = 1\}$.
2. Under a linear model for controls, $E(Y | Z = 0, X) = \beta_0 + \beta_X^\top X$, obtain the OLS-based ATT estimator.
3. Show that

$$\text{att} = E(Y | Z = 1) - E\left\{ \frac{e(X)(1-Z)Y}{e\{1-e(X)\}} \right\},$$

where $e(X)$ is the propensity score and $e = \Pr(Z = 1)$; derive the IPW estimator.

4. Derive the doubly robust ATT estimator and show double robustness.

Answer

首先

$$\text{ATT} = E(Y | Z = 1) - E\{Y(0) | Z = 1\}.$$

由 $Y(0) \perp\!\!\!\perp Z | X$ 与 consistency,

$$E\{Y(0) | Z = 1\} = E\{E(Y | Z = 0, X) | Z = 1\}.$$

故

$$\text{ATT} = E(Y | Z = 1) - E\{E(Y | Z = 0, X) | Z = 1\}.$$

若控制组 outcome model 为 $E(Y | Z = 0, X) = \beta_0 + \beta_X^\top X$, 则

$$\widehat{\text{ATT}}_{\text{or}} = \bar{Y}_1 - \frac{1}{n_1} \sum_{Z_i=1} (\hat{\beta}_0 + \hat{\beta}_X^\top X_i).$$

令 $e = \Pr(Z = 1)$, $e(X) = \Pr(Z = 1 | X)$ 。由 Bayes 公式,

$$E\{Y(0) | Z = 1\} = \frac{1}{e} E\left\{\frac{e(X)(1-Z)Y}{1-e(X)}\right\}.$$

因此

$$\widehat{\text{ATT}}_{\text{ipw}} = \bar{Y}_1 - \frac{1}{\hat{e}n} \sum_i \frac{\hat{e}(X_i)(1-Z_i)Y_i}{1-\hat{e}(X_i)}.$$

DR 形式可写成

$$\widehat{\text{ATT}}_{\text{dr}} = \bar{Y}_1 - \frac{1}{n_1} \sum_i \left[Z_i \hat{\mu}_0(X_i) + \frac{\hat{e}(X_i)(1-Z_i)}{1-\hat{e}(X_i)} \{Y_i - \hat{\mu}_0(X_i)\} \right].$$

若 $\hat{e}(X)$ 或 $\hat{\mu}_0(X)$ 至少一个模型正确, 则该估计量一致。

题 18 — PS3-3: ATT 估计量模拟

来源: Problem Set 3, Problem 3。

原题:

设计模拟比较 ATT 的 OR、IPW 与 DR 估计量。

1. Describe the goal of the simulation study.
2. Describe the data generating mechanism.
3. Display simulation results, including at least bias, variance, and bootstrap variance estimator; other properties such as asymptotic normality and CI coverage may also be reported.
4. Explain the results.
5. (选做) Investigate whether, for ATT, IPW using an estimated propensity score can outperform IPW using the true propensity score, as sometimes occurs for ACE.

Answer

一个清楚的模拟目标是比较三类估计量在 outcome model 错设、propensity model 错设、两者都正确时的 bias、variance、bootstrap SE 与置信区间覆盖率。示例设计:

$$X \sim N(0, I_p), \quad Z | X \sim \text{Bernoulli}\{\text{expit}(\alpha_0 + \alpha^\top X)\},$$

$$Y(0) = m_0(X) + \varepsilon, \quad Y(1) = Y(0) + \tau(X),$$

其中可令 $m_0(X)$ 和 $\tau(X)$ 包含非线性项, 用来制造模型错设。每次模拟记录真实

$$\text{ATT} = E\{\tau(X) | Z = 1\}$$

以及三种估计。

报告表格至少包含: Monte Carlo bias、Monte Carlo SD、平均 bootstrap SE、RMSE、95% CI coverage。若 propensity score 接近 0 或 1, IPW 方差会显著变大; DR 在一个 nuisance model 正确时通常更稳定。关于“真 propensity score 与估计 propensity score”的选做现象, ACE 中估计 propensity score 有时更优, ATT 中也可能发生类似现象, 但依赖 estimand 权重与模型设定, 必须用模拟结果而不是口头断言。

第 4 次作业

题 19 — Ding 13.1: τ_T, τ_C, τ 的关系

来源: 教材 Chapter 13, Problem 13.1; 附件 HW4; 已有 TeX: `homework4.tex`。

原题:

在 $Z \perp\!\!\!\perp \{Y(1), Y(0)\} \mid X$ 下, 证明

$$\tau_T - \tau = \frac{\text{cov}\{e(X), \tau(X)\}}{e}, \quad \tau_C - \tau = -\frac{\text{cov}\{e(X), \tau(X)\}}{1 - e}.$$

Here $e(X) = \Pr(Z = 1 \mid X)$, $e = \Pr(Z = 1)$, and $\tau(X) = E\{Y(1) - Y(0) \mid X\}$. The problem asks how ATT, ATC, and ACE differ when treatment selection is related to treatment-effect heterogeneity.

Answer

由 ignorability,

$$\tau_T = E\{\tau(X) \mid Z = 1\} = \frac{E\{\tau(X)e(X)\}}{e}.$$

又

$$E\{\tau(X)e(X)\} = \text{cov}\{e(X), \tau(X)\} + e\tau.$$

所以

$$\tau_T = \tau + \frac{\text{cov}\{e(X), \tau(X)\}}{e}.$$

同理

$$\tau_C = E\{\tau(X) \mid Z = 0\} = \frac{E[\tau(X)\{1 - e(X)\}]}{1 - e} = \tau - \frac{\text{cov}\{e(X), \tau(X)\}}{1 - e}.$$

因此差异完全由 treatment selection 与 effect heterogeneity 的协方差控制。

题 20 — Ding 13.6 与 13.10: individual effect predictor 和一般 estimand 的 DR

来源: 教材 Chapter 13, Problems 13.6, 13.10; 附件 HW4。

原题:

1. Ding 13.6. Assume IID units $\{Z_i, X_i, Y_i(1), Y_i(0)\}_{i=1}^n$. The individual effect is $\tau_i = Y_i(1) - Y_i(0)$ and the CATE is $\tau(X_i) = E\{Y_i(1) - Y_i(0) \mid X_i\}$. Under randomization $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\}$ with $e = \Pr(Z_i = 1)$, show that

$$\delta_i = \frac{Z_i Y_i}{e} - \frac{(1 - Z_i) Y_i}{1 - e}$$

is an unbiased predictor of the individual effect: $E(\delta_i - \tau_i) = 0$, and $E(\delta_i) = \tau$. Under ignorability given X_i , replace e by $e(X_i)$ and show $E(\delta_i - \tau_i) = 0$ and $E\{\delta_i - \tau(X_i)\} = 0$.

2. Ding 13.10. For a given $h(X)$, define DR components

$$\begin{aligned}\tilde{\mu}_1^{h,\text{dr}} &= E \left[\frac{Zh(X)\{Y - \mu_1(X, \beta_1)\}}{e(X, \alpha)} + h(X)\mu_1(X, \beta_1) \right], \\ \tilde{\mu}_0^{h,\text{dr}} &= E \left[\frac{(1-Z)h(X)\{Y - \mu_0(X, \beta_0)\}}{1 - e(X, \alpha)} + h(X)\mu_0(X, \beta_0) \right].\end{aligned}$$

Show the corresponding double robustness for estimating $E\{h(X)Y(1)\}$, $E\{h(X)Y(0)\}$, and the weighted estimand τ^h .

Answer

随机化下,

$$\delta_i = \frac{Z_i Y_i}{e} - \frac{(1 - Z_i) Y_i}{1 - e} = \frac{Z_i Y_i(1)}{e} - \frac{(1 - Z_i) Y_i(0)}{1 - e}.$$

条件于潜在结果,

$$E(\delta_i | Y_i(1), Y_i(0)) = Y_i(1) - Y_i(0) = \tau_i.$$

所以 $E(\delta_i - \tau_i) = 0$ 且 $E(\delta_i) = \tau$ 。Ignorability 下把 e 换成 $e(X_i)$, 再条件于 $X_i, Y_i(1), Y_i(0)$ 即得同样结论, 并且 $E(\delta_i | X_i) = \tau(X_i)$ 。

对一般 estimand

$$\tau^h = \frac{E[h(X)\{Y(1) - Y(0)\}]}{E\{h(X)\}},$$

DR 形式为

$$\begin{aligned}\tilde{\mu}_1^{h,\text{dr}} &= E \left[\frac{Zh(X)\{Y - \mu_1^*(X)\}}{e^*(X)} + h(X)\mu_1^*(X) \right], \\ \tilde{\mu}_0^{h,\text{dr}} &= E \left[\frac{(1-Z)h(X)\{Y - \mu_0^*(X)\}}{1 - e^*(X)} + h(X)\mu_0^*(X) \right].\end{aligned}$$

若 propensity score 正确, 或相应 outcome model 正确, 则 $\tilde{\mu}_z^{h,\text{dr}} = E\{h(X)Y(z)\}$; 因此比值给出 τ^h 。

题 21 — Ding 14.3: 二元结果的 weighted logistic regression

来源: 教材 Chapter 14, Problem 14.3; 附件 HW4。

原题:

With a binary outcome, replace linear outcome models with logistic outcome models. Show that, when the logistic regressions use the corresponding causal weights and include intercepts, the doubly robust estimators equal the outcome-regression estimators. The result should be shown for both τ and τ_T .

Answer

以 treatment arm 为例, weighted logistic regression 的 score equation 含有截距项:

$$\sum_i \frac{Z_i h(X_i)}{e^*(X_i)} \{Y_i - \mu_1^*(X_i)\} = 0.$$

这正是 DR 估计中 augmentation residual 项的样本和。因此

$$\hat{\mu}_1^{h,dr} = n^{-1} \sum_i h(X_i) \mu_1^*(X_i).$$

Control arm 同理。也就是说, 若 outcome model 通过相同权重拟合, 并包含截距, DR 公式中的残差校正项由估计方程强制为零, 所以 DR 数值上退化为 outcome regression estimator。该结论同时适用于 ATE 与 ATT 对应的 $h(X)$ 。

题 22 — Ding 17.1, 17.2, 17.6 与 Ding 20.1

来源: 教材 Chapter 17, Problems 17.1, 17.2, 17.6; Chapter 20, Problem 20.1; 附件 HW4。

原题:

1. Ding 17.1. Show that $f(x) = (ax + 1)/(bx + 1)$ is increasing in x if $a > b$ and decreasing in x if $a < b$.
2. Ding 17.2. Revisit the proof of the E-value theorem. Assume $Z \perp\!\!\!\perp Y \mid (X, U)$. Show that if $\mathbb{R}_{ZU|x} > 1$ but $\mathbb{R}_{UY|x} < 1$, then $\mathbb{R}_{ZY|x}^{\text{obs}} < 1$.
3. Ding 17.6. For binary Z, Y, U and implicit conditioning on X , assume latent ignorability $Z \perp\!\!\!\perp Y \mid U$. Show

$$\text{RD}_{ZY}^{\text{obs}} = \text{RD}_{ZU} \times \text{RD}_{UY}.$$

Then derive the Cornfield-type inequalities for risk differences:

$$\min(\text{RD}_{ZU}, \text{RD}_{UY}) \geq \text{RD}_{ZY}^{\text{obs}}, \quad \max(\text{RD}_{ZU}, \text{RD}_{UY}) \geq \sqrt{\text{RD}_{ZY}^{\text{obs}}}.$$

4. Ding 20.1. Under strict overlap $\eta \leq e(X) \leq 1 - \eta$, show that $\eta \leq e \leq 1 - \eta$ and prove the textbook covariate-balance implication bounding the average absolute difference in covariate means between treatment groups in terms of p , e , η , and the largest eigenvalues of $\text{cov}(X \mid Z = 1)$ and $\text{cov}(X \mid Z = 0)$.

Answer

Chapter 17 的问题围绕 bounding factor theorem。若观测 risk ratio $RR_{ZY|x}^{\text{obs}} > 1$, 且未观测混杂 U 与 treatment、outcome 的最大关联强度为 $RR_{ZU|x}$ 和 $RR_{UY|x}$, 则 bounding factor 形式为

$$RR_{ZY|x}^{\text{obs}} \leq \frac{RR_{ZU|x} RR_{UY|x}}{RR_{ZU|x} + RR_{UY|x} - 1}.$$

17.1 与 17.2 主要补齐这个上界的代数单调性和技术条件; 17.6 则把同类思想转到 risk difference。解题时要明确: 这些不等式不是估计模型, 而是说明要解释掉一个大的 observed association, 未观测混杂必须同时强烈关联 treatment 与 outcome。

Chapter 20.1 讨论 overlap 权重。若 $h(X) = e(X)\{1 - e(X)\}$, 则对应目标总体强调 overlap 区域, 权重在 $e(X)$ 接近 0 或 1 时自动变小。证明通常从一般加权 estimand

$$\tau^h = \frac{E\{h(X)\tau(X)\}}{E\{h(X)\}}$$

出发, 再代入 $h(X) = e(X)\{1 - e(X)\}$ 推出 treated 和 control 的样本权重分别正比于

$1 - e(X)$ 与 $e(X)$ 。

题 23 — PS4-2: sensitivity analysis 的识别公式

来源: Problem Set 4, Problem 2。

原题:

设

$$\frac{E\{Y(1) \mid Z = 1, X\}}{E\{Y(1) \mid Z = 0, X\}} = \varepsilon_1(X), \quad \frac{E\{Y(0) \mid Z = 1, X\}}{E\{Y(0) \mid Z = 0, X\}} = \varepsilon_0(X).$$

推导 outcome-regression、IPW 与 DR 型识别公式。

1. Show that

$$\begin{aligned} E\{Y(1) \mid Z = 0\} &= E\{\mu_1(X)/\varepsilon_1(X) \mid Z = 0\}, \\ E\{Y(0) \mid Z = 1\} &= E\{\mu_0(X)\varepsilon_0(X) \mid Z = 1\}, \end{aligned}$$

and hence write the ACE identification formula.

2. Show the IPW identities

$$E\{Y(1)\} = E\left\{w_1(X) \frac{ZY}{e(X)}\right\}, \quad E\{Y(0)\} = E\left\{w_0(X) \frac{(1-Z)Y}{1-e(X)}\right\},$$

where $w_1(X) = e(X) + \{1 - e(X)\}/\varepsilon_1(X)$ and $w_0(X) = e(X)\varepsilon_0(X) + \{1 - e(X)\}$.

3. (选做) Derive a doubly robust estimation formula.

Answer

记 $\mu_z(X) = E(Y \mid Z = z, X)$ 。由 consistency,

$$\mu_1(X) = E\{Y(1) \mid Z = 1, X\}, \quad \mu_0(X) = E\{Y(0) \mid Z = 0, X\}.$$

因此

$$\begin{aligned} E\{Y(1) \mid Z = 0, X\} &= \mu_1(X)/\varepsilon_1(X), \\ E\{Y(0) \mid Z = 1, X\} &= \mu_0(X)\varepsilon_0(X). \end{aligned}$$

对 $X \mid Z$ 积分可得

$$\begin{aligned} E\{Y(1) \mid Z = 0\} &= E\{\mu_1(X)/\varepsilon_1(X) \mid Z = 0\}, \\ E\{Y(0) \mid Z = 1\} &= E\{\mu_0(X)\varepsilon_0(X) \mid Z = 1\}. \end{aligned}$$

所以

$$\text{ACE} = E\{ZY + (1 - Z)\mu_1(X)/\varepsilon_1(X)\} - E\{Z\mu_0(X)\varepsilon_0(X) + (1 - Z)Y\}.$$

IPW 形式来自对 X 条件化:

$$\begin{aligned} E\{Y(1)\} &= E\left[\left\{e(X) + \frac{1 - e(X)}{\varepsilon_1(X)}\right\} \frac{ZY}{e(X)}\right], \\ E\{Y(0)\} &= E\left[\{e(X)\varepsilon_0(X) + 1 - e(X)\} \frac{(1 - Z)Y}{1 - e(X)}\right]. \end{aligned}$$

DR 公式则把上述 outcome-regression 和 IPW 残差校正合并：若 $\mu_z(X)$ 或 $e(X)$ 至少一侧模型正确，对应项仍保持一致。写作时应说明 $\varepsilon_z(X)$ 是 sensitivity parameter，而不是数据可检验量。

第 5 次作业

题 24 — Ding 21.1, 21.6, 21.7: IV 与 CACE

来源: 教材 Chapter 21, Problems 21.1, 21.6, 21.7; 附件 HW5; 已有 TeX: `homework5.tex`。

原题:

1. Ding 21.1. For the Wald estimator

$$\hat{\tau}_c = \frac{\hat{\tau}_Y}{\hat{\tau}_D}, \quad \hat{\tau}_Y = \bar{Y}_1 - \bar{Y}_0, \quad \hat{\tau}_D = \bar{D}_1 - \bar{D}_0,$$

show that its finite-sample variance can be infinite:

$$\text{var}(\hat{\tau}_c) = \infty.$$

2. Ding 21.6. Consider a binary IV Z , an ordinal treatment received $D \in \{0, 1, \dots, J\}$, and potential outcomes $Y(z, d)$. Under IV randomization, monotonicity $D(1) \geq D(0)$, and exclusion restriction $Y(z, d) = Y(d)$, prove

$$\frac{E(Y | Z = 1) - E(Y | Z = 0)}{E(D | Z = 1) - E(D | Z = 0)} = \sum_{j=1}^J w_j E\{Y(j) - Y(j-1) | D(1) \geq j > D(0)\},$$

where

$$w_j = \frac{\Pr\{D(1) \geq j > D(0)\}}{\sum_{j'=1}^J \Pr\{D(1) \geq j' > D(0)\}}.$$

3. Ding 21.7. Analyze the flu-shot encouragement design data `fludata.txt`. Variables include `assign` (encouragement), `receive` (flu shot), `outcome` (flu-related hospitalization), and covariates such as age, sex, race, COPD, diabetes, heart disease, renal disease, and liver disease. Analyze the data with and without covariate adjustment; report first stage, ITT, CACE, uncertainty, and identifying assumptions.

Answer

Wald 估计量为

$$\hat{\tau}_c = \frac{\hat{\tau}_Y}{\hat{\tau}_D}.$$

有限样本中 $\hat{\tau}_D = 0$ 有正概率, 所以 ratio estimator 的二阶矩可发散, Ding 21.1 的结论是

$$\text{var}(\hat{\tau}_c) = \infty.$$

对 ordinal treatment $D \in \{0, 1, \dots, J\}$, 在 randomization、monotonicity $D(1) \geq D(0)$

与 exclusion restriction $Y(z, d) = Y(d)$ 下,

$$\frac{E(Y | Z = 1) - E(Y | Z = 0)}{E(D | Z = 1) - E(D | Z = 0)} = \sum_{j=1}^J w_j E\{Y(j) - Y(j-1) | D(1) \geq j > D(0)\},$$

其中

$$w_j = \frac{\Pr\{D(1) \geq j > D(0)\}}{\sum_{l=1}^J \Pr\{D(1) \geq l > D(0)\}}.$$

证明使用分解

$$D(1) - D(0) = \sum_{j=1}^J \mathbb{1}\{D(1) \geq j > D(0)\}$$

以及 $Y(D(1)) - Y(D(0))$ 的相邻差分展开。

Flu shot 数据本地包含 `fludata.txt`, 已有分析给出:

$$\hat{\tau}_Y = -0.0147, \quad \hat{\tau}_D = 0.1184, \quad \hat{\tau}_c = -0.1246.$$

Lin 调整后约为

$$\hat{\tau}_{Y,L} = -0.0149, \quad \hat{\tau}_{D,L} = 0.1187, \quad \hat{\tau}_{c,L} = -0.1252.$$

解释为: 在随机鼓励、单调性、排除限制成立时, 鼓励诱导接种的人群中, 接种流感疫苗使 flu-related hospitalization 概率约下降 12.5 个百分点。

题 25 — Ding 22.2 与 22.7: compliers 的风险比与整体 ACE bounds

来源: 教材 Chapter 22, Problems 22.2, 22.7; 附件 HW5。

原题:

1. Ding 22.2. With a binary outcome, define the complier risk ratio

$$RR_c = \frac{\Pr\{Y(1) = 1 | U = c\}}{\Pr\{Y(0) = 1 | U = c\}}.$$

Under the standard IV assumptions, show

$$RR_c = \frac{E(DY | Z = 1) - E(DY | Z = 0)}{E\{(D-1)Y | Z = 1\} - E\{(D-1)Y | Z = 0\}}.$$

2. Ding 22.7. Let $Y(d)$ be the potential outcome under treatment received d , and define the whole-population treatment-received ACE

$$\delta = E\{Y(d=1) - Y(d=0)\}.$$

With principal strata $u \in \{a, n, c\}$ and $m_{du} = E\{Y(d) | U = u\}$, note that m_{0a} and m_{1n} are not identified. If $Y \in [\underline{y}, \bar{y}]$, prove the lower and upper bounds for δ in terms of identified quantities and the outcome support.

Answer

令

$$\pi_c = E(D | Z = 1) - E(D | Z = 0).$$

在标准 IV 假设下,

$$\mu_{1c} = \pi_c^{-1} \{E(DY | Z = 1) - E(DY | Z = 0)\},$$

$$\mu_{0c} = \pi_c^{-1} [E\{(1 - D)Y | Z = 0\} - E\{(1 - D)Y | Z = 1\}].$$

由于 $(D - 1)Y = -(1 - D)Y$,

$$\frac{\mu_{1c}}{\mu_{0c}} = \frac{E(DY | Z = 1) - E(DY | Z = 0)}{E\{(D - 1)Y | Z = 1\} - E\{(D - 1)Y | Z = 0\}}.$$

对整体 treatment-received effect

$$\delta = E\{Y(d = 1) - Y(d = 0)\},$$

always-takers 的 m_{0a} 与 never-takers 的 m_{1n} 不可识别。若 $Y \in [\underline{y}, \bar{y}]$, 则

$$\delta = \delta' + \pi_n m_{1n} - \pi_a m_{0a}.$$

取不可识别项的最小/最大可能值, 得到

$$\underline{\delta} = \delta' - \bar{y} \Pr(D = 1 | Z = 0) + \underline{y} \Pr(D = 0 | Z = 1),$$

$$\bar{\delta} = \delta' - \underline{y} \Pr(D = 1 | Z = 0) + \bar{y} \Pr(D = 0 | Z = 1).$$

题 26 — Ding 23.1, 23.2 (选做), 23.3: TSLS 代数

来源: 教材 Chapter 23, Problems 23.1–23.3; 附件 HW5。

原题:

1. Ding 23.1. In TSLS, the first-stage fitted treatment satisfies $\hat{D}_i = \hat{\Gamma}^\top Z_i$ with

$$\hat{\Gamma} = \left(\sum_i Z_i Z_i^\top \right)^{-1} \sum_i Z_i D_i^\top.$$

Show that the TSLS estimator

$$\hat{\beta}_{\text{tsls}} = \left(\sum_i \hat{D}_i \hat{D}_i^\top \right)^{-1} \sum_i \hat{D}_i Y_i$$

reduces to the just-identified IV estimator when Z and D have the same dimension and the relevant sample moment matrices are invertible.

2. Ding 23.2 (选做). With a single endogenous treatment and a single IV, prove that the indirect least squares estimator equals TSLS. In the structural and reduced-form equations, show

$$\hat{\beta}_{1,\text{ils}} = \frac{\hat{\Gamma}_1}{\hat{\gamma}_1} = \hat{\beta}_{1,\text{tsls}}.$$

3. Ding 23.3. Define the control-function estimator by first regressing D on Z to obtain residuals $\check{D}_i$, then regressing Y on D and $\check{D}$. Show that the coefficient of D equals $\hat{\beta}_{\text{tsls}}$.

Answer

第一阶段把 D_i 投影到 Z_i :

$$\hat{\Gamma} = \left(\sum_i Z_i Z_i^\top \right)^{-1} \sum_i Z_i D_i^\top, \quad \hat{D}_i = \hat{\Gamma}^\top Z_i.$$

第二阶段

$$\hat{\beta}_{\text{TSLs}} = \left(\sum_i \hat{D}_i \hat{D}_i^\top \right)^{-1} \sum_i \hat{D}_i Y_i.$$

若 Z 与 D 同维且矩阵可逆, 代入 $\hat{D}_i = \hat{\Gamma}^\top Z_i$ 并消去公共矩阵, 可得

$$\hat{\beta}_{\text{TSLs}} = \left(\sum_i Z_i D_i^\top \right)^{-1} \sum_i Z_i Y_i = \hat{\beta}_{\text{IV}}.$$

单一 IV 与 treatment 时, ILS 为 reduced form 除以 first stage:

$$\hat{\beta}_{\text{ILs}} = \frac{\widehat{\text{cov}}(Z, Y)}{\widehat{\text{cov}}(Z, D)}.$$

而 TSLs 中 $\hat{D} = \hat{\gamma}_0 + \hat{\gamma}_1 Z$, 所以

$$\frac{\widehat{\text{cov}}(\hat{D}, Y)}{\widehat{\text{var}}(\hat{D})} = \frac{\hat{\gamma}_1 \widehat{\text{cov}}(Z, Y)}{\hat{\gamma}_1^2 \widehat{\text{var}}(Z)} = \frac{\widehat{\text{cov}}(Z, Y)}{\widehat{\text{cov}}(Z, D)}.$$

Control-function 回归把第一阶段残差 $\check{D} = D - \hat{D}$ 放入 outcome regression。由 FWL 定理, 控制 $\check{D}$ 后 D 的剩余部分正是 $\hat{D}$, 因此其系数等于 TSLs。

题 27 — PS5-2: Wald 公式的协方差形式

来源: Problem Set 5, Problem 2。

原题:

证明

$$\text{cace} = \frac{\text{cov}(Z, Y)}{\text{cov}(Z, D)}.$$

The problem starts from the CACE identification formula under randomization and asks for the equivalent covariance form when Z is binary.

Answer

二元 Z 下, 对任意变量 A ,

$$\text{cov}(Z, A) = \Pr(Z = 1) \Pr(Z = 0) \{E(A | Z = 1) - E(A | Z = 0)\}.$$

分别取 $A = Y$ 与 $A = D$, 公共因子 $\Pr(Z = 1)\Pr(Z = 0)$ 抵消, 得到

$$\frac{\text{cov}(Z, Y)}{\text{cov}(Z, D)} = \frac{E(Y | Z = 1) - E(Y | Z = 0)}{E(D | Z = 1) - E(D | Z = 0)}.$$

在 randomization、monotonicity、exclusion restriction 下, 右侧就是 CACE。

题 28 — PS5-3 (选做): 观测研究中 CACE 的 weighting method

来源: Problem Set 5, Problem 3.

原题:

在 $Z \perp\!\!\!\perp \{Y(1), Y(0), D(1), D(0)\} | X$ 下证明 κ_0, κ_1 公式。Specifically, monotonicity and exclusion restriction continue to hold, while randomization of Z is replaced by conditional ignorability. Define

$$\begin{aligned}\kappa_{0i} &= (1 - D_i) \frac{(1 - Z_i) - \Pr(Z_i = 0 | X_i)}{\Pr(Z_i = 0 | X_i) \Pr(Z_i = 1 | X_i)}, \\ \kappa_{1i} &= D_i \frac{Z_i - \Pr(Z_i = 1 | X_i)}{\Pr(Z_i = 0 | X_i) \Pr(Z_i = 1 | X_i)}.\end{aligned}$$

Show that

$$\begin{aligned}E\{g(Y_i(0), X_i) | U_i = c\} &= \frac{1}{\Pr(U_i = c)} E\{\kappa_{0i} g(Y_i, X_i)\}, \\ E\{g(Y_i(1), X_i) | U_i = c\} &= \frac{1}{\Pr(U_i = c)} E\{\kappa_{1i} g(Y_i, X_i)\}.\end{aligned}$$

Answer

令 $e(X) = \Pr(Z = 1 | X)$ 。题中

$$\kappa_1 = D \frac{Z - e(X)}{e(X)\{1 - e(X)\}}, \quad \kappa_0 = (1 - D) \frac{(1 - Z) - \{1 - e(X)\}}{e(X)\{1 - e(X)\}}.$$

以 κ_1 为例, 对 X 条件化, 并用 consistency:

$$E\{\kappa_1 g(Y, X) | X\} = E\left[D \frac{Z - e(X)}{e(X)\{1 - e(X)\}} g(Y, X) | X\right].$$

展开 $Z = 1$ 和 $Z = 0$ 两项, 可化为

$$E\{[D(1) - D(0)]g(Y(1), X) | X\}.$$

在 monotonicity 下 $D(1) - D(0) = \mathbb{1}(U = c)$, 于是

$$E\{\kappa_1 g(Y, X)\} = E\{\mathbb{1}(U = c)g(Y(1), X)\} = \Pr(U = c)E\{g(Y(1), X) | U = c\}.$$

κ_0 的证明相同, 只是替换为 $Y(0)$ 。

题 29 — PS5-4: Card (1993) return to schooling 数据分析

来源: Problem Set 5, Problem 4; 数据文件 `card.csv` 未在本仓库中找到。

原题:

Card (1993) used geographic variation in college proximity to estimate the return to schooling. The treatment is education, the outcome is log wage, and the IV is college proximity. The file `card.csv` is a modified version of the data from Card's homepage; variable explanations are in `code_bk.txt`. The analysis must state causal quantities, assumptions, model choices, and interpretation.

1. Dichotomize education and infer CACE for units with $D(1) = 1, D(0) = 0$, where D is the dichotomized education variable. Explain the threshold choice, sensitivity to the threshold, and conduct an analysis with covariates.
2. Without dichotomizing education, use standard TSLS. Compare the results with the dichotomized analysis.

Answer

若把 education 二值化为 $D = \mathbb{1}(\text{educ} \geq c)$, college proximity 为 IV Z , 则估计的是阈值 c 所定义的 complier 群体的 schooling-threshold effect。阈值 c 应有实质解释, 例如高中毕业、上大学、大学毕业等, 并应做敏感性分析:

$$\hat{\tau}_c(c) = \frac{\bar{Y}_{Z=1} - \bar{Y}_{Z=0}}{\bar{D}_{Z=1} - \bar{D}_{Z=0}}.$$

加入协变量时, 可分别对 Y 和 D 做 Lin/OLS 调整后取 ratio, 或使用饱和/线性 first stage。

不二值化 education 时, 标准 TSLS 模型为

$$\text{educ}_i = \gamma_0 + \gamma_1 Z_i + \gamma_X^\top X_i + u_i,$$

$$\log(\text{wage}_i) = \beta_0 + \beta_1 \widehat{\text{educ}_i} + \beta_X^\top X_i + \varepsilon_i.$$

此时 β_1 是由 proximity 诱导的教育年限边际变化的 LATE 型解释, 而不是某个阈值下的 CACE。报告中必须写清 instrument relevance、exclusion restriction、monotonicity/linear IV interpretation, 并比较不同阈值、不同协变量集合和连续 TSLS 的结果。

第 6 次作业

题 30 — Ding 26.1–26.3: principal score 方法

来源: 教材 Chapter 26, Problems 26.1–26.3; 附件 HW6; 已有 TeX: `homework6.tex`。

原题:

1. Ding 26.1. Under strong monotonicity $M(0) = 0$, complete the proof for the principal-stratum effect $\tau(0, 0)$. In particular, prove the identification formula for units with $M(1) = 0, M(0) = 0$ using the principal score $\pi(X) = \Pr\{M(1) = 1 \mid X\}$ and principal ignorability.
2. Ding 26.2. Replace strong monotonicity with ordinary monotonicity $M(1) \geq M(0)$. Define principal scores

$$\pi_{(m_1, m_0)}(X) = \Pr\{M(1) = m_1, M(0) = m_0 \mid X\}.$$

Under the two principal ignorability conditions for $Y(1)$ and $Y(0)$ across relevant strata, derive the principal score identification formulas for $\tau(1, 0)$, $\tau(0, 0)$, and $\tau(1, 1)$.

3. Ding 26.3. Replace complete randomization by observational-study ignorability

$$Z \perp\!\!\!\perp \{M(1), M(0), Y(1), Y(0)\} \mid X,$$

with positivity $0 < e(X) < 1$. Under strong monotonicity and principal ignorability, prove the IPW principal-score identification formulas for $\tau(1, 0)$ and $\tau(0, 0)$.

Answer

在 strong monotonicity $M(0) = 0$ 下, $\pi(X) = \Pr\{M(1) = 1 \mid X\}$ 可由

$$\pi(X) = \Pr(M = 1 \mid Z = 1, X)$$

识别。若还满足 principal ignorability

$$E\{Y(0) \mid M(1) = 1, X\} = E\{Y(0) \mid M(1) = 0, X\},$$

则

$$\tau(1, 0) = E(Y \mid Z = 1, M = 1) - \frac{E\{\pi(X)Y \mid Z = 0\}}{\pi},$$

$$\tau(0, 0) = E(Y \mid Z = 1, M = 0) - \frac{E[\{1 - \pi(X)\}Y \mid Z = 0]}{1 - \pi}.$$

第二个式子的证明: $(Z = 1, M = 0)$ 等于 stratum $M(1) = 0, M(0) = 0$, 给出 treated potential outcome; control potential outcome 由 principal ignorability 加权恢复。

Ordinary monotonicity $M(1) \geq M(0)$ 下, 可能 strata 为 $(1, 1), (1, 0), (0, 0)$:

$$\pi_{00}(X) = \Pr(M = 0 \mid Z = 1, X), \quad \pi_{11}(X) = \Pr(M = 1 \mid Z = 0, X),$$

$$\pi_{10}(X) = \Pr(M = 1 \mid Z = 1, X) - \Pr(M = 1 \mid Z = 0, X).$$

观测研究中再加入 treatment propensity score $e(X)$ 和 positivity, 使用 IPW 把 $Z = 1$ 与 $Z = 0$ 的条件分布还原到目标总体; 其余结构与 randomized case 平行。

题 31 — Ding 27.2, 27.8 与 Ding 28.2

来源: 教材 Chapter 27, Problems 27.2, 27.8; Chapter 28, Problem 28.2; 附件 HW6。

原题:

1. Ding 27.2. Let $Y(z, m)$ be the potential outcome and $M(z)$ the potential mediator. Show that the two sequential-randomization assumptions

$$Z \perp\!\!\!\perp Y(z, m) \mid X, \quad M \perp\!\!\!\perp Y(z, m) \mid (Z, X)$$

are equivalent, under positivity, to joint randomization

$$(Z, M) \perp\!\!\!\perp Y(z, m) \mid X.$$

2. Ding 27.8. Suppose

$$E(M \mid Z, X) = \beta_0 + \beta_1 Z + \beta_2^\top X,$$

$$E(Y \mid Z, M, X) = \theta_0 + \theta_1 Z + \theta_2 M + \theta_3 ZM + \theta_4^\top X.$$

Show

$$\text{nde} = \theta_1 + \theta_3\{\beta_0 + \beta_2^\top E(X)\}, \quad \text{nie} = (\theta_2 + \theta_3)\beta_1,$$

and describe estimation with IID data.

3. Ding 28.2. Let $Z \in \{1, \dots, K\}$ be a multi-valued treatment. Under

$$Z \perp\!\!\!\perp \{Y(1), \dots, Y(K)\} \mid X, \quad e_k(X) = \Pr(Z = k \mid X) > 0,$$

define $\mu_k = E\{Y(k)\}$ and $\mu_k(X) = E(Y \mid Z = k, X)$. Prove outcome-regression, IPW, and doubly robust formulas for μ_k .

Answer

Sequential randomization 常写为

$$Z \perp\!\!\!\perp \{Y(z, m), M(z) : z, m\} \mid X,$$

$$M \perp\!\!\!\perp \{Y(z, m) : z, m\} \mid Z, X.$$

它等价于一组 joint randomization 条件: 先给 treatment 分配可忽略性, 再给 mediator 在给定 (Z, X) 后可忽略性。27.2 的证明本质是反复使用条件独立的分解和 consistency, 把不可观测的 cross-world 均值转写成观测条件均值。

若 outcome model 含 treatment-mediator interaction, 例如

$$E(Y \mid Z, M, X) = \theta_0 + \theta_1 Z + \theta_2 M + \theta_3 ZM + \theta_X^\top X,$$

则 direct 与 indirect effects 不再简单等于单个主效应系数。以固定 mediator m 的 controlled direct effect 为例:

$$\text{CDE}(m) = E\{Y(1, m) - Y(0, m)\} = \theta_1 + \theta_3 m.$$

对多值 treatment 的 observational studies, 28.2 使用广义 propensity score。若

$$Z \perp\!\!\!\perp \{Y(z, m) : z, m\} \mid X$$

且 overlap 成立, 则

$$E\{Y(z, m)\} = E\{E(Y | Z = z, M = m, X)\}$$

在相应 mediator 条件也可识别时成立。报告时应区分 CDE、NDE、NIE 对 cross-world assumptions 的要求。

题 32 — PS6-2: one-sided noncompliance 下无排除限制的 CACE bounds

来源: Problem Set 6, Problem 2。

原题:

在 $D(0) = 0$ 、strong monotonicity 下, 二元 outcome 且不使用 exclusion restriction, 推导 CACE bounds。The original problem considers an encouragement experiment with one-sided noncompliance, $D_i(0) = 0$ for all i . Under strong monotonicity and exclusion restriction CACE is point-identified, but the task here is to derive bounds on CACE for a binary outcome without exclusion restriction.

Answer

One-sided noncompliance 下 $D(0) = 0$, 且 $D(1) \in \{0, 1\}$ 。Compliers 为 $D(1) = 1$, never-takers 为 $D(1) = 0$ 。随机化识别

$$\pi_c = \Pr(D = 1 | Z = 1).$$

对 compliers, 在 $Z = 1, D = 1$ 中可直接观测 $Y(1)$, 所以

$$E\{Y(1) | U = c\} = E(Y | Z = 1, D = 1).$$

但不使用 exclusion restriction 时, $Z = 0$ 下所有人都 $D = 0$, 其中混合了 compliers 和 never-takers, 不能点识别 $E\{Y(0) | U = c\}$ 。二元 outcome 给出

$$0 \leq E\{Y(0) | U = c\} \leq 1.$$

更紧地, 令 $p_0 = E(Y | Z = 0)$, 则

$$p_0 = \pi_c E\{Y(0) | U = c\} + (1 - \pi_c) E\{Y(0) | U = n\}.$$

由于 never-taker 均值在 $[0, 1]$,

$$\max \left\{ 0, \frac{p_0 - (1 - \pi_c)}{\pi_c} \right\} \leq E\{Y(0) | U = c\} \leq \min \left\{ 1, \frac{p_0}{\pi_c} \right\}.$$

因此 CACE 的 bounds 为

$$E(Y | Z = 1, D = 1) - U_0 \leq \text{CACE} \leq E(Y | Z = 1, D = 1) - L_0,$$

其中 L_0, U_0 分别是上式中 control potential outcome mean 的下界和上界。

题 33 — PS6-3: individual causal effect 与 probabilities of causation

来源: Problem Set 6, Problem 3; Tian and Pearl (2000), Mueller and Pearl (2022)。

原题:

令 $Y(1), Y(0)$ 均为二元, 比较 ace 与 $\tau = \Pr\{Y(1) > Y(0)\}$, 并给出 bounds。

1. Explain the difference between ace and τ in a real context such as a new drug and survival. Give a numerical example with $\text{ace} = 0$ but $\tau > 0$.
2. Let $\pi_{y_1 y_0} = \Pr\{Y(1) = y_1, Y(0) = y_0\}$. Show $\text{ace} = \pi_{10} - \pi_{01}$ and $\tau = \pi_{10}$.
3. Derive bounds on τ .
 - (a) In randomized trials, express $\pi_{11}, \pi_{00}, \pi_{01}$ in terms of π_{10} and the observed marginal risks, then solve $0 \leq \pi_{y_1 y_0} \leq 1$.
 - (b) In non-randomized studies without ignorability, show the weak bound

$$0 \leq \pi_{10} \leq \Pr(Z = 1, Y = 1) + \Pr(Z = 0, Y = 0).$$

Answer

定义 principal strata 比例

$$\pi_{y_1 y_0} = \Pr\{Y(1) = y_1, Y(0) = y_0\}.$$

则

$$\text{ACE} = E\{Y(1) - Y(0)\} = \pi_{10} - \pi_{01}, \quad \tau = \Pr\{Y(1) > Y(0)\} = \pi_{10}.$$

所以 ACE 是净改善率, 而 τ 是被 treatment 真正救助/改善的比例。可有 $\text{ACE} = 0$ 但 $\tau > 0$: 例如 $\pi_{10} = \pi_{01} = 0.25$, 其余 0.5 放在 $(1, 1)$ 或 $(0, 0)$ 。

随机实验中, 记

$$p_1 = \Pr(Y = 1 | Z = 1) = \Pr\{Y(1) = 1\}, \quad p_0 = \Pr(Y = 1 | Z = 0) = \Pr\{Y(0) = 1\}.$$

由

$$\pi_{11} = p_1 - \pi_{10}, \quad \pi_{00} = 1 - p_0 - \pi_{10}, \quad \pi_{01} = \pi_{10} - (p_1 - p_0),$$

并要求四个 π 非负, 得到

$$\max(0, p_1 - p_0) \leq \tau = \pi_{10} \leq \min(p_1, 1 - p_0).$$

非随机研究中, 不假设 ignorability 时, 题目给出的弱上界为

$$0 \leq \pi_{10} \leq \Pr(Z = 1, Y = 1) + \Pr(Z = 0, Y = 0),$$

因为 π_{10} 可分解为 $Z = 1$ 与 $Z = 0$ 两部分, 而这两部分分别被可观测的 $(Z = 1, Y = 1)$ 与 $(Z = 0, Y = 0)$ 覆盖。

题 34 — PS6-4: principal stratification 与 direct/indirect effects

来源: Problem Set 6, Problem 4; VanderWeele (2008)。

原题:

证明若个体 NDE 或 CDE 为零, 则某些 principal strata 中 principal causal effect 为零。Let Z_i be binary treatment, M_i a binary post-treatment variable, and Y_i the outcome. Define individual direct effects

$$\text{cde}_i(m) = Y_i(1, m) - Y_i(0, m), \quad \text{nde}_i(z) = Y_i(1, M_i(z)) - Y_i(0, M_i(z)).$$

1. Show that if $\text{nde}_i(0) = 0$ for all i , then the principal causal effect is zero for strata $M(1) = M(0) = m$, $m = 0, 1$.
2. Show that if $\text{cde}_i(m) = 0$ for all i and $m = 0, 1$, then the principal causal effect is zero for strata $M(1) = M(0) = m$, $m = 0, 1$.

Answer

若 $M(1) = M(0) = m$, 则 stratum 内

$$Y(1) - Y(0) = Y(1, M(1)) - Y(0, M(0)) = Y(1, m) - Y(0, m) = \text{CDE}_i(m).$$

因此如果 $\text{CDE}_i(m) = 0$ 对所有 i, m 成立, 则在 $(M(1), M(0)) = (m, m)$ strata 中

$$E\{Y(1) - Y(0) \mid M(1) = M(0) = m\} = 0.$$

若 $\text{NDE}_i(0) = Y_i(1, M_i(0)) - Y_i(0, M_i(0)) = 0$ 对所有 i 成立, 则对 $M(1) = M(0) = m$ 的单位, $M_i(0) = m$, 所以

$$Y_i(1, m) - Y_i(0, m) = 0.$$

于是同样推出 $(0, 0)$ 与 $(1, 1)$ 两个不改变 mediator 的 principal strata 中 principal causal effect 为零。

题 35 — PS6-5: media framing, anxiety mediation analysis

来源: Problem Set 6, Problem 5; 本地已有分析脚本与结果表。

原题:

Brader, Valentino, and Suhay (2008) conducted a randomized experiment in the United States on media stories about immigration. The news story manipulated both benefit-versus-cost framing and whether the pictured immigrant was white from Russia or Hispanic from Mexico. To simplify, define $Z = 1$ as the cost frame with a Hispanic immigrant picture. The mediator is anxiety, measured on an approximately continuous scale, and the outcome is a binary indicator for agreeing to send a letter about immigration policy. The data are available as **framing** in the R package **mediation**.

1. Define the natural direct and indirect effects and state the identification assumptions.
2. Estimate NDE and NIE, and compute their variances and 95% confidence intervals.
3. Interpret the results.

Answer

令 treatment $Z = 1$ 表示 cost frame 加 Hispanic immigrant picture, mediator M 为 anxiety, outcome Y 为是否同意写信。自然直接效应和自然间接效应可写为

$$\text{NDE} = E\{Y(1, M(0)) - Y(0, M(0))\},$$

$$\text{NIE} = E\{Y(1, M(1)) - Y(1, M(0))\}.$$

识别通常需要: treatment 随机化或给定协变量后无 treatment-outcome / treatment-mediator confounding; 给定 (Z, X) 后无 mediator-outcome confounding; 以及相应 positivity 和 consistency。

本地结果文件给出的估计为:

Effect	Estimate	Variance	SE	2.5%	97.5%
NDE	0.0346	0.00367	0.0606	-0.0837	0.1591
NIE	0.0674	0.00053	0.0229	0.0254	0.1184
Total effect	0.1021	0.00441	0.0664	-0.0293	0.2278

解释: 自然间接效应为正且区间不含 0, 说明 anxiety channel 有证据支持; 自然直接效应估计为正但不精确, 区间覆盖 0。因此 framing 对写信意愿的影响主要表现为通过 anxiety 的间接路径。